WASP-52b

R-0322

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_221122` · created 2026-07-18T02:26:16+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459905.6589 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.014
Transit depth (Rp/Rs)^2	1.95 +/- 0.4 [%]
Orbital Inclination (inc)	89.85 +/- 1.5
Airmass coefficient 1 (a1)	0.959 +/- 0.012
Airmass coefficient 2 (a2)	0.033 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	#2 - [303, 247]
Optimal Aperture	4.44
Optimal Annulus	7.4
Transit Duration (day)	0.0852 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.14 ± 0.014 vs 0.1646 (deviation 1.76σ)
Duration fit vs published	0.0852 d vs 0.0758 d (deviation 4.68σ)
Mid-time O–C	-12.1 min
Depth SNR	10.0
Residual scatter	1.28 %

Light curve

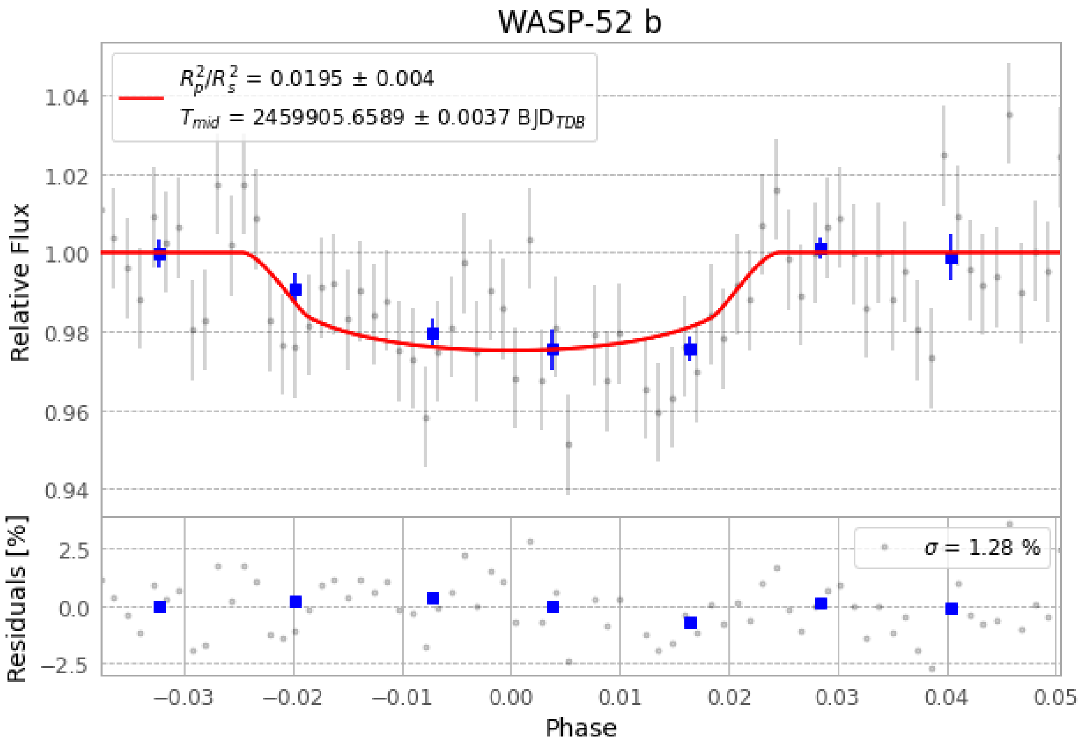

Figure 1 — FinalLightCurve_WASP-52 b_2022-11-21.png (SHA-256 0433a67bf5e21d55…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 176], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 0b4ac79f98f87dcd…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c`

Data provenance

75 FITS frames (49 MB), `WASP-52221122020915.FITS` → `WASP-52221122055115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221122.log (f68460cacf47…), FinalLightCurve_WASP-52 b_2022-11-21.png (0433a67bf5e2…), FinalLightCurve_WASP-52 b_2022-11-21.csv (13650cafc96c…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-18 02:26Z.